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a. The APT Pricing Equation

According to the Arbitrage Pricing Theory, for a **well-diversified portfolio** w, the idiosyncratic risk ε_w is diversified away (approaching zero), leaving only the systematic risk. The equation linking the expected return to its risk profile is:

$$E(\tilde{r}_w) = r_f + \beta_w[E(\tilde{r}_M) - r_f]$$

b. APT vs. CAPM Security Market Line

While the APT equation and the CAPM's SML are algebraically identical in form, they differ in scope: the **CAPM** is an equilibrium theory that claims the relationship holds for **all individual assets**, whereas the **APT** is a no-arbitrage theory that only guarantees this relationship for **well-diversified portfolios** (and "most" individual assets).
Furthermore, the APT arrives at this result without requiring the restrictive CAPM assumptions of quadratic utility or normally distributed returns.

c. Arbitrage Strategy and the Value of w

To make the trading strategy **risk-free, self-financing, and profitable**, a "tracking portfolio" that mimics the systematic risk of target portfolio must be constructed. Since the tracking portfolio consists of a share w in the market portfolio (where $\beta_M = 1$) and $1 - w$ in the risk-free asset (where $\beta_{rf} = 0$), the beta of the tracking portfolio is simply w.

To neutralize the market risk and ensure the strategy is risk-free, set:

$$\mathbf{w = \beta_w}$$

Why this works:

- **Risk-Free:** By setting $w = \beta_w$, the exposure to the market factor $[\tilde{r}_M - E(\tilde{r}_M)]$ in the long position is exactly offset by the short position, and since the portfolio is well-diversified, the idiosyncratic risk is negligible.
- **Self-Financing:** Selling the tracking portfolio provides the exact amount of capital needed to buy the mispriced portfolio at $t = 0$.
- **Profitable for Sure:** At $t = 1$, the net payoff will be the "alpha" or the deviation Δ, which is positive by assumption.

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a. Expected Return for Portfolio One

For any well-diversified portfolio w with $\beta_{w,m} = \beta_{w,v} = 0$, the APT requires that its idiosyncratic risk vanishes, so its return is purely deterministic and equal to the risk-free rate:

$$E(\tilde{r}_w^1) = r_f.$$

b. Expected Returns for Portfolios Two and Three

For the second well-diversified portfolio with $\beta_{w,m} = 1$ and $\beta_{w,v} = 0$ (the mimicking portfolio for the market factor), the APT pricing relation implies

$$E(\tilde{r}_w^2) = r_f + [E(\tilde{r}_M) - r_f] = E(\tilde{r}_M).$$

For the third well-diversified portfolio with $\beta_{w,m} = 0$ and $\beta_{w,v} = 1$ (the mimicking portfolio for the value factor), the APT pricing relation implies

$$E(\tilde{r}_w^3) = r_f + [E(\tilde{r}_V) - r_f].$$

c. Trading Strategy for Portfolio Four ($\Delta < 0$)

The APT requires that *any* well-diversified portfolio with the *same* factor loadings $\beta_{w,m}$ and $\beta_{w,v}$ must have the *same* expected return

$$E(\tilde{r}_w) = r_f + \beta_{w,m}[E(\tilde{r}_M) - r_f] + \beta_{w,v}[E(\tilde{r}_V)].$$

Portfolio 4 has exactly these loadings but offers a strictly *lower* expected return (by the amount $|\Delta|$). Therefore portfolio 4 is overpriced relative to the fair APT price.

Construct a synthetic portfolio 5 that replicates *exactly* the same betas as portfolio 4 by taking:

- weight $1 - \beta_{w,m} - \beta_{w,v}$ in the zero-beta risk-free portfolio (portfolio 1 from part (a)),
- weight $\beta_{w,m}$ in the market-mimicking portfolio (portfolio 2),
- weight $\beta_{w,v}$ in the value-mimicking portfolio (portfolio 3).

Portfolio 5 is well-diversified, has identical factor exposures to portfolio 4, and therefore *must* have the higher (fair) expected return given by the APT formula above.

Arbitrage trading strategy (self-financing, zero net risk, positive sure profit):

- Go long $x > 0$ dollars in portfolio 5.
- Go short x dollars in portfolio 4.

Net investment at $t = 0$: zero (self-financing).
Net factor exposures: zero (the betas cancel).
Net idiosyncratic risk: zero (both portfolios are well-diversified).
Net payoff at $t = 1$: $x \times (-\Delta) > 0$ with certainty (the difference in expected returns, with all risky components exactly offsetting).

Because $\Delta < 0$, this strategy earns a riskless profit with no capital outlay, violating the no-arbitrage assumption of the APT. Investors would immediately exploit it, driving the price of portfolio 4 up (and its expected return down) until $\Delta = 0$ and the APT pricing equation is restored.